

Massachusetts Health Network Database

DAMG 6210: Data Management and Database Design

GitHub Repository: <https://github.com/harpereskey/DAMG6210>

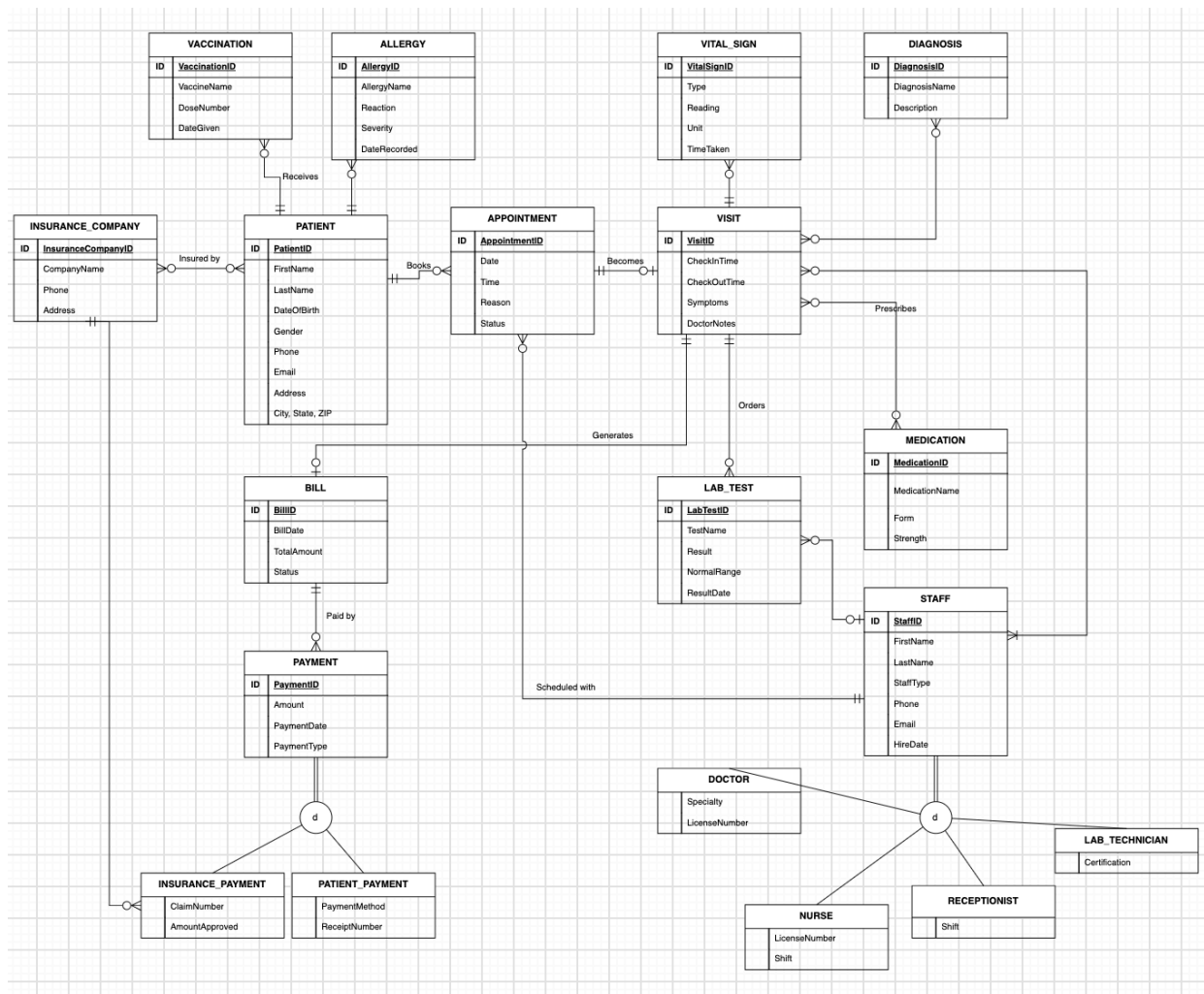
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Revised Scope/Mission Statement: The Massachusetts Health Network Database is a database that maintains patient care and billing records for one outpatient healthcare facility in Massachusetts. The database contains patient information such as demographic information, insurance information, allergies, and vaccination history. Patients can also book appointments with staff members. When an appointment becomes a visit, the database records check-in and check-out times, symptoms, doctor's notes, vital signs, diagnoses, medications prescribed, and lab tests ordered during the visit. Lab tests are performed by staff members, typically lab technicians, who record the results in the system.

Staff is divided into four different roles: doctors, nurses, receptionists, and lab technicians. Staff is the main supertype, and each role has its own specific information. For example, doctors have their specialty and license number, nurses have their license number and shift, receptionists have their shift details, and lab technicians have their certification. After a patient's visit, a bill is created, which the patient can pay either in one payment or through multiple payments.

Revised Mission Objectives:

1. **Patient Record Management:** Maintain accurate patient information, including demographics, insurance information, allergies, and vaccination history.
2. **Appointment Scheduling:** Schedule and track appointments between patients and staff, including appointment dates, times, reasons, and status.
3. **Patient Visit Management:** Track completed patient visits, including check-in and check-out times, symptoms, doctor's notes, and the appointment associated with each visit.
4. **Staff Management:** Maintain information about doctors, nurses, receptionists, and lab technicians, including their role-specific qualifications, licenses, certifications, specialties, etc.
5. **Treatment and Clinical Tracking:** Record vital signs, diagnoses, and medications prescribed during patient visits to maintain an organized history of patient care.
6. **Lab Test Management:** Track lab tests ordered during patient visits, including test information, results, and the lab technicians responsible for performing them.
7. **Billing and Payment Processing:** Generate bills associated with patient visits and track one or multiple payments made toward each bill.
8. **Insurance Management:** Maintain patient insurance information and track payments made by insurance companies toward patient bills.



Business Problems:

1. Patient information needs to be organized: Patient demographics, insurance, allergies, and vaccination history need to be stored together so staff can easily access important information.
2. Appointments and visits need to be tracked: Patients may cancel or miss appointments, so the facility needs to track appointment status and know which appointments actually become visits.
3. Patient care needs to be documented: Information such as vital signs, diagnoses, medications, and lab tests needs to be recorded and connected to the correct patient visit.

4. Different staff roles need to be managed: The facility has doctors, nurses, receptionists, and lab technicians with different responsibilities and information that needs to be stored.
5. Lab tests need to be tracked: Lab tests need to stay connected to the visit where they were ordered, along with their results and the staff involved.
6. Bills and payments need to be managed: A bill may have multiple payments and can be paid by both the patient and an insurance company, so each payment needs to be tracked.
7. Insurance information needs to be connected: Patient insurance information needs to be stored and insurance companies need to be connected to the payments they make.

Entities and their relationships:

- Patient: Books appointments, receives vaccinations, has allergies, and is insured by one or more insurance companies.
- Appointment: Booked by a patient, scheduled with a staff member, and may become a visit.
- Visit: Created from an appointment; records vital signs and diagnoses, includes prescribed medications and ordered lab tests, and generates a bill.
- Vaccination: Received by a patient when needed.
- Allergy: Associated with a patient, if applicable.
- Vital Sign: Recorded during a patient's visit.
- Diagnosis: Recorded during or after a patient's visit.
- Medication: Prescribed to the patient during a visit.
- Lab Test: Ordered during a visit and performed by a staff member.
- Staff: Involved in appointments and visits; includes doctors, nurses, receptionists, and lab technicians.
- Doctor: Type of staff member at the facility, with a specialty and license number.
- Nurse: Type of staff member at the facility, with a license number and shift.
- Receptionist: Type of staff member at the facility, with shift details.
- Lab Technician: Type of staff member at the facility, with a certification.
- Bill: Generated from a visit and paid through one or more payments.
- Payment: Associated with a bill and can be either an insurance payment or a patient payment.
- Insurance Payment: Type of payment made by an insurance company.
- Patient Payment: Type of payment made directly by the patient.
- Insurance Company: Insures patients and makes insurance payments.

Key design decisions:

- One outpatient facility: We decided to focus on one outpatient healthcare facility to keep the scope manageable. Multiple locations, hospital stays, emergency care, and surgeries were not included.
- Appointments and visits are separate: We separated them because not every appointment becomes a visit. A patient may cancel or not show up, while a visit represents the actual care received.
- Patient medical information is separated: Allergies and vaccinations have their own entities because a patient can have multiple allergies and vaccinations.
- Visit information is separated: Vital signs, diagnoses, medications, and lab tests have their own entities because a single visit can have multiple records of each.
- Staff has different roles: Staff is the main supertype because all employees share basic information. Doctor, Nurse, Receptionist, and Lab Technician are separate subtypes because each role also has its own specific information.
- Lab tests are connected to visits and staff: Lab tests are connected to the visit where they were ordered and the staff involved in performing them.
- Bills and payments are separate: We separated them because one bill can have multiple payments. This allows part of a bill to be paid by insurance and the rest by the patient.
- There are two types of payments: Payment is the main entity, while Insurance Payment and Patient Payment are subtypes because each type has different information.
- Insurance companies are separate: A patient can be insured by more than one insurance company (for example, primary and secondary coverage), and an insurance company can insure many patients and make many insurance payments.